High-Order Implicit Manufactured FDA Solver Verification Report

Generated report for `highOrderManufacturedFDAImplicit`

May 2, 2026

Abstract

This report documents the manufactured-solution finite-volume solver `highOrderManufacturedFDAImplicit`. The solver advances the same manufactured FDA activation problem as the explicit code, but uses a high-order implicit matrix discretization for V_m and a theta-type implicit update for the state variables. The convergence figures are generated by `plot_convergence.m` from the data in `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/highOrderManufacturedFDAImplicit/r`

1 Manufactured FDA problem

The exact solution and reaction terms are the same as in the explicit solver. The reported two-dimensional exact fields are

$$F(x,y) = \cos(\pi x)\cos(2\pi y), \qquad G(x,y) = 1 + xy^2, \tag{1}$$

$$V_m^\star = \sqrt{1+t}\,F, \qquad u_1^\star = (1+t)G + V_m^\star, \tag{2}$$

$$u_2^\star = \frac{1}{(1+t)\sqrt{G}}, \qquad u_3^\star = 0. \tag{3}$$

The parameter β is computed from the conductivity tensor and the manufactured Laplacian mode so that the exact fields satisfy the coupled PDE–ODE system.

2 High-order matrix discretization

The implicit solver assembles a high-order mass matrix M and stiffness matrix K once during setup for the fixed mesh and constant conductivity. The mass matrix can be `lumped`, i.e. diagonal, or `consistent`, where the LRE reconstruction is integrated over the cell quadrature points. The stiffness matrix is assembled from LRE face-gradient coefficients and face quadrature. The current implementation requires `useHighOrder_Vm=true`; the state source terms may use high-order LRE reconstruction through `useHighOrder_states`.

After scaling by $1/(\chi C_m)$ and separating the manufactured linear beta term, the discrete potential operator is

$$L = \frac{1}{\chi C_m}K - \frac{\beta}{\chi C_m}M. \tag{4}$$

The expected spatial orders used in the plots are 2 for `p1`, 3 for `p2`, and 4 for `p3`.

3 Implicit time integration

The time integration uses a theta method, with

$$\theta = 1 \quad \text{for backward Euler}, \qquad \theta = \frac{1}{2} \quad \text{for Crank–Nicolson}. \tag{5}$$

The potential solve has the form

$$\left(\frac{M}{\Delta t} - \theta L\right) V_m^{n+1} = \left(\frac{M}{\Delta t} + (1-\theta)L\right) V_m^n + \theta(S^{n+1} + b^{n+1}) + (1-\theta)(S^n + b^n), \tag{6}$$

where S contains the nonlinear source contribution and b contains the manufactured Dirichlet boundary contribution. Nonlinear coupling is handled by a fixed number of outer corrections, and the states are updated by a theta-consistent local implicit iteration. The linear system is solved with Eigen using either `SparseLU` or `BiCGSTAB`, according to `implicitHOCoeffs`.

4 Error measures and convergence rates

The reported errors are the same as in the explicit solver: cell-centred volume-weighted L_1, L_2, L_∞, and Gauss-reconstructed norms. Pairwise rates are computed with consecutive mesh sizes and $h_N = 1/N$. The summary table reports the all-mesh log–log fit of the cell-centred L_2 errors.

5 Observed orders

Table 1: Implicit solver: observed cell-centred L_2 convergence orders. The columns labelled "all" are log–log fits over all available meshes, while the columns labelled $N \geq 80$ are fits over the finest meshes only. Here bE denotes backward Euler, cN denotes Crank–Nicolson, cons. denotes a consistent mass matrix, and lump. denotes a lumped mass matrix.

V_m interp.	State interp.	Scheme	Mass	Δt	V_m all	u_1 all	u_2 all	V_m $N \geq 80$	u_1 $N \geq 80$	u_2 $N \geq 80$
p1	p1	bE	cons.	10^{-3}	1.996	1.574	0.265	2.625	1.613	0.012
p1	p1	bE	cons.	10^{-4}	1.995	1.700	1.357	2.613	2.411	0.781
p1	p1	bE	lump.	10^{-3}	1.996	1.574	0.265	2.625	1.613	0.012
p1	p1	bE	lump.	10^{-4}	1.995	1.700	1.357	2.613	2.411	0.781
p1	p1	cN	cons.	10^{-3}	1.994	1.703	1.712	2.611	2.435	2.440
p1	p1	cN	cons.	10^{-4}	1.994	1.703	1.712	2.611	2.435	2.440
p1	p1	cN	lump.	10^{-3}	1.994	1.703	1.712	2.611	2.435	2.440
p1	p1	cN	lump.	10^{-4}	1.994	1.703	1.712	2.611	2.435	2.440
p2	p2	bE	cons.	10^{-3}	2.270	1.214	0.092	2.313	0.240	0.000
p2	p2	bE	cons.	10^{-4}	2.199	2.035	0.821	2.052	1.428	0.018
p2	p2	bE	lump.	10^{-3}	2.216	1.232	0.095	2.268	0.273	0.000
p2	p2	bE	lump.	10^{-4}	2.154	2.009	0.844	2.043	1.489	0.023
p2	p2	cN	cons.	10^{-3}	2.192	2.213	2.212	2.026	2.031	2.025
p2	p2	cN	cons.	10^{-4}	2.192	2.213	2.213	2.026	2.031	2.031
p2	p2	cN	lump.	10^{-3}	2.148	2.166	2.165	2.021	2.024	2.020
p2	p2	cN	lump.	10^{-4}	2.148	2.166	2.166	2.021	2.024	2.025
p3	p3	bE	cons.	10^{-3}	2.355	0.402	0.004	0.036	0.001	0.000
p3	p3	bE	cons.	10^{-4}	3.465	1.530	0.177	2.597	0.019	0.000
p3	p3	bE	lump.	10^{-3}	3.335	0.495	0.006	5.464	0.031	-0.000
p3	p3	bE	lump.	10^{-4}	2.437	1.556	0.213	2.239	0.342	0.000
p3	p3	cN	cons.	10^{-3}	3.630	3.378	2.847	4.370	4.119	0.798
p3	p3	cN	cons.	10^{-4}	3.630	3.381	3.392	4.371	4.150	4.152
p3	p3	cN	lump.	10^{-3}	2.382	2.414	2.371	2.041	2.085	1.818
p3	p3	cN	lump.	10^{-4}	2.382	2.413	2.413	2.042	2.080	2.077

6 Convergence figures

Run `plot_convergence.m` from this report directory to create the PDF and PNG figures. Each plot compares bE (backward Euler) and cN (Crank–Nicolson) for both mass-matrix choices at a fixed time step.

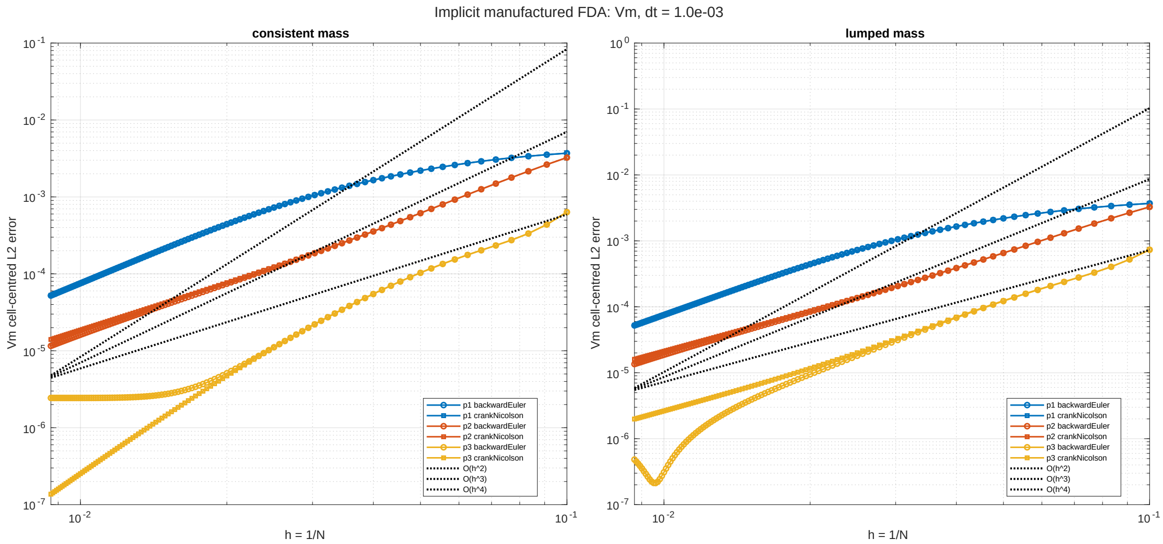

Figure 1: Implicit solver, V_m cell-centred L_2 error for $\Delta t = 10^{-3}$. Each figure compares bE and cN for consistent and lumped mass matrices. Dotted guide lines show $O(h^2)$, $O(h^3)$, and $O(h^4)$.

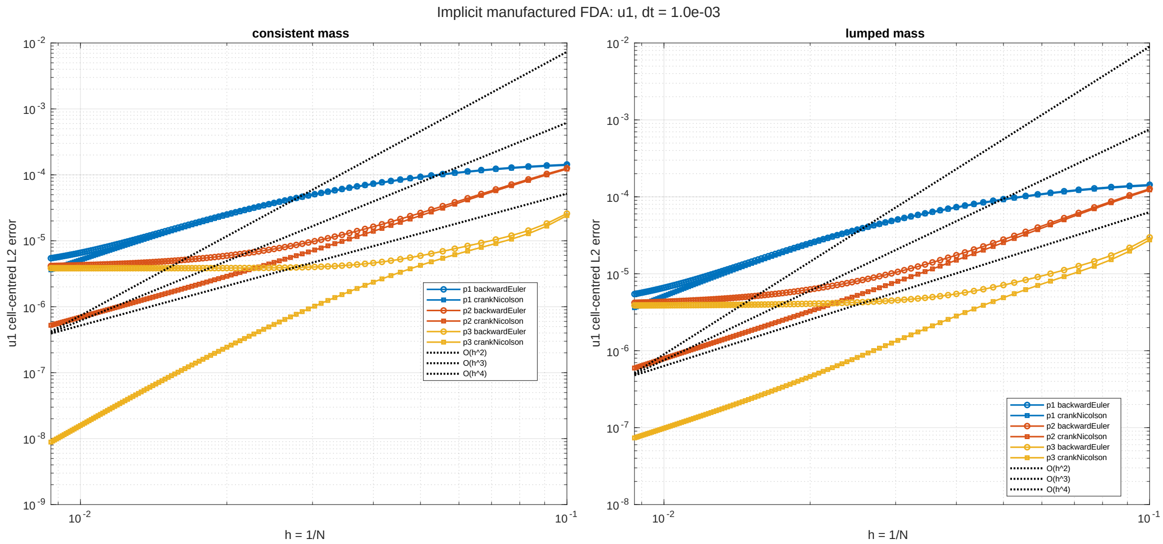

Figure 2: Implicit solver, u_1 cell-centred L_2 error for $\Delta t = 10^{-3}$. Each figure compares bE and cN for consistent and lumped mass matrices. Dotted guide lines show $O(h^2)$, $O(h^3)$, and $O(h^4)$.

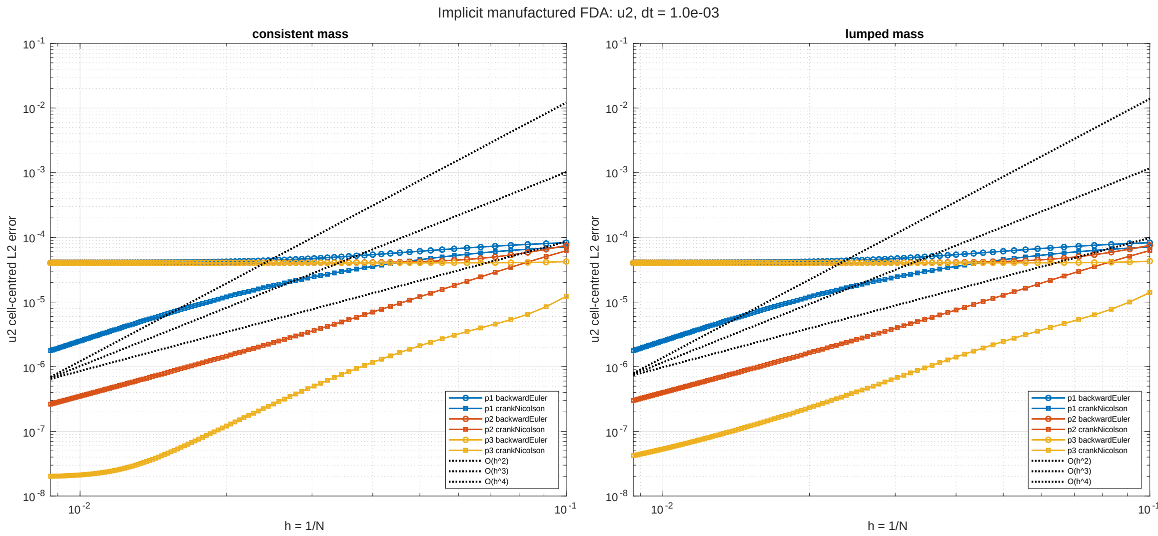

Figure 3: Implicit solver, u_2 cell-centred L_2 error for $\Delta t = 10^{-3}$. Each figure compares bE and cN for consistent and lumped mass matrices. Dotted guide lines show $O(h^2)$, $O(h^3)$, and $O(h^4)$.

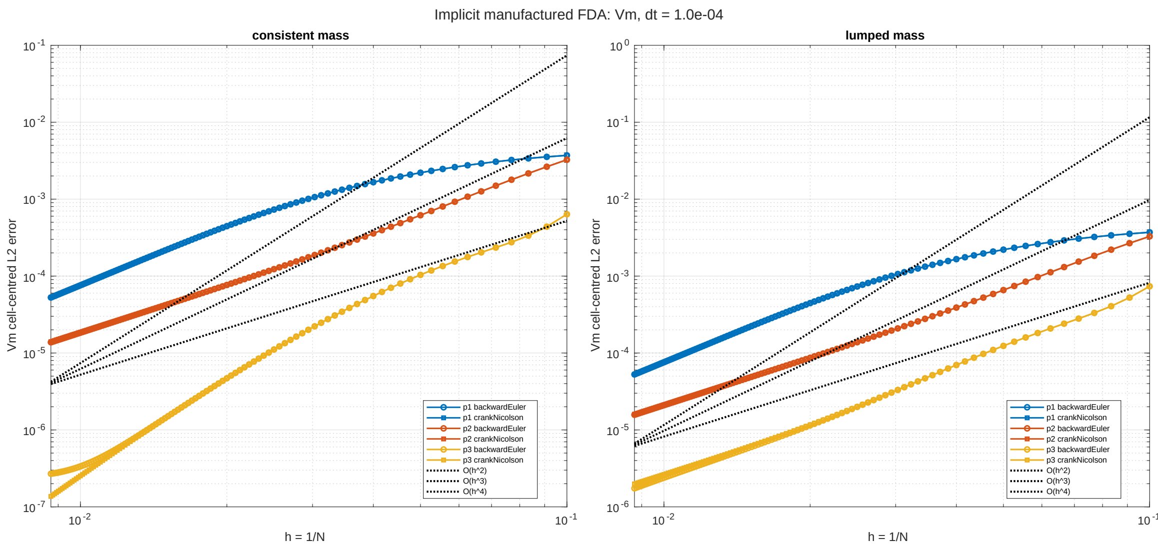

Figure 4: Implicit solver, V_m cell-centred L_2 error for $\Delta t = 10^{-4}$. Each figure compares bE and cN for consistent and lumped mass matrices. Dotted guide lines show $O(h^2)$, $O(h^3)$, and $O(h^4)$.

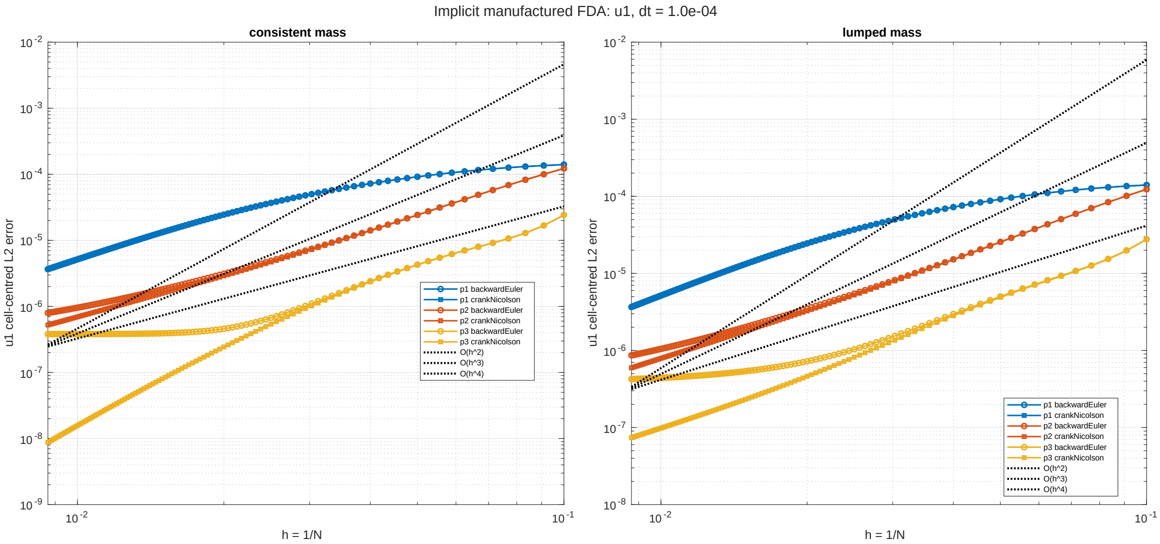

Figure 5: Implicit solver, u_1 cell-centred L_2 error for $\Delta t = 10^{-4}$. Each figure compares bE and cN for consistent and lumped mass matrices. Dotted guide lines show $O(h^2)$, $O(h^3)$, and $O(h^4)$.

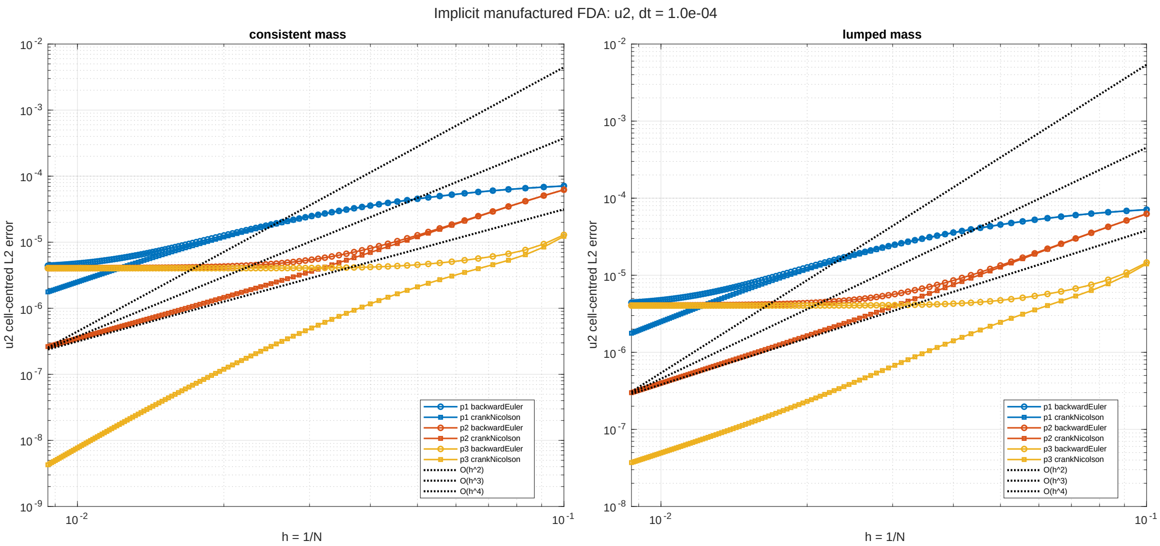

Figure 6: Implicit solver, u_2 cell-centred L_2 error for $\Delta t = 10^{-4}$. Each figure compares bE and cN for consistent and lumped mass matrices. Dotted guide lines show $O(h^2)$, $O(h^3)$, and $O(h^4)$.

7 Notes

For the same spatial order, Crank–Nicolson should reduce the temporal contribution relative to backward Euler when the time step is not already negligible. The mass-matrix comparison isolates the effect of lumped versus consistent integration in the high-order potential solve.